EXPERIMENT 6

```
clc;

 clear; close all;

% Parameters Am = 1; Ac = 1; fm = 5; fc = 50; beta = 5; kp = 5; fs = 1000;

t = 0:1/fs:1;

% Message Signal m = Amcos(2pifmt);

% Carrier Signal c = Accos(2pifct);

% Frequency Modulated Signal fm_sig = Accos(2pifct + betasin(2pifmt));

% Phase Modulated Signal pm_sig = Accos(2pifct + kp*m);

% Frequency Deviation and Modulation Index Df = beta*fm;

fprintf('Frequency Deviation = %.2f Hz\n', Df); fprintf('FM Modulation Index = %.2f\n', beta);

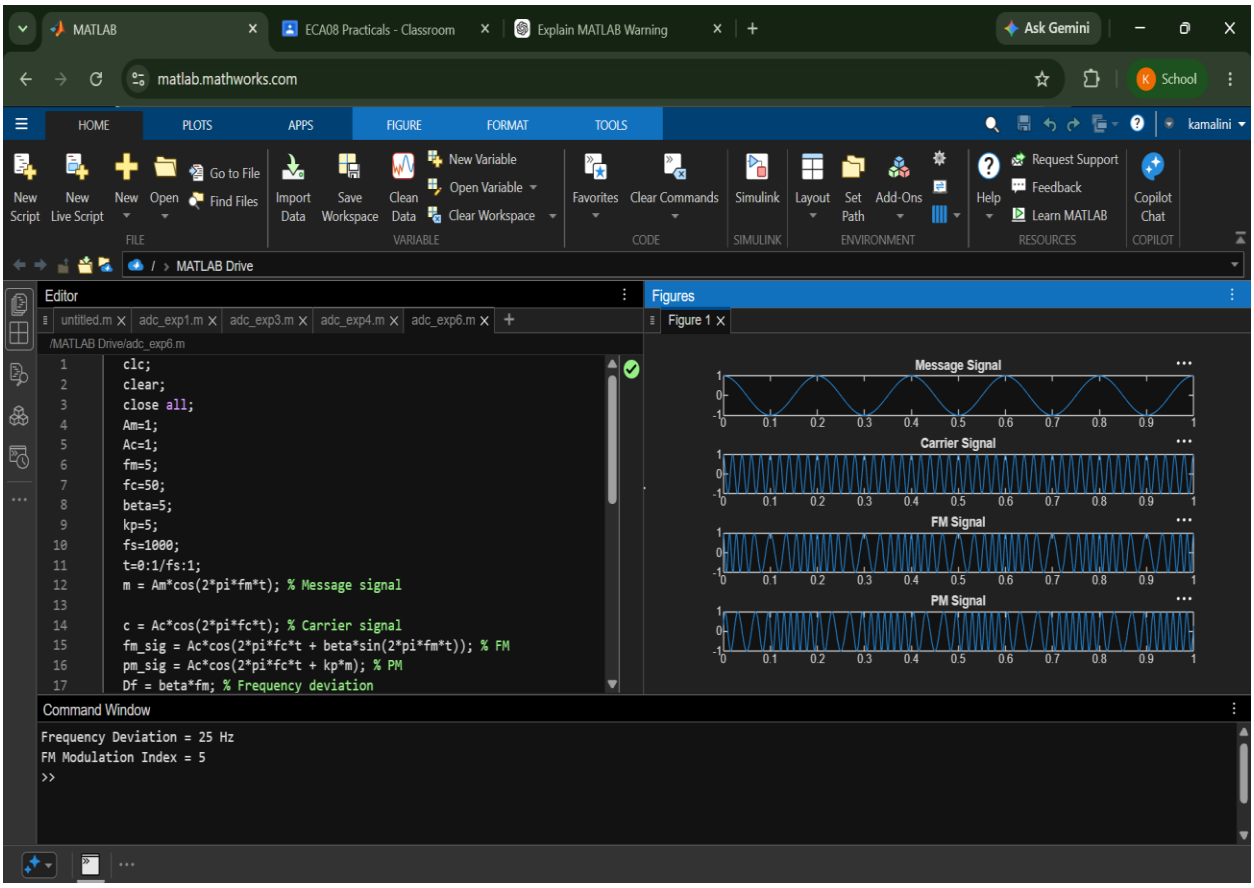
% Plot Signals figure;

subplot(4,1,1); plot(t, m); title('Message Signal'); xlabel('Time (s)'); ylabel('Amplitude'); grid on;

subplot(4,1,2); plot(t, c); title('Carrier Signal'); xlabel('Time (s)'); ylabel('Amplitude'); grid on;

subplot(4,1,3); plot(t, fm_sig); title('Frequency Modulated (FM) Signal'); xlabel('Time (s)');
ylabel('Amplitude'); grid on;

subplot(4,1,4); plot(t, pm_sig); title('Phase Modulated (PM) Signal'); xlabel('Time (s)');
ylabel('Amplitude'); grid on;
```


MATLAB
ECA08 Practicals - Classroom
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Import Data
Save Workspace
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New Variable
Open Variable
Clear Workspace
Favorites
Clear Commands
Simulink
Layout
Set Path
Add-Ons
Help
Request Support
Feedback
Learn MATLAB
Copilot Chat
FILE
VARIABLE
CODE
SIMULINK
ENVIRONMENT
RESOURCES
COPILOT
MATLAB Drive
Editor
untitled.m
adc_exp1.m
adc_exp3.m
adc_exp4.m
adc_exp6.m
/MATLAB Drive/adc_exp6.m
clc;
clear;
close all;
Am=1;
Ac=1;
fm=5;
fc=50;
beta=5;
kp=5;
fs=1000;
t=0:1/fs:1;
m = Am*cos(2*pi*fm*t); % Message signal
c = Ac*cos(2*pi*fc*t); % Carrier signal
fm_sig = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t)); % FM
pm_sig = Ac*cos(2*pi*fc*t + kp*m); % PM
Df = beta*fm; % Frequency deviation
Figures
Figure 1
Message Signal
Carrier Signal
FM Signal
PM Signal
Command Window
Frequency Deviation = 25 Hz
FM Modulation Index = 5
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